

Prelab 13: AC Circuits - Phase Relationships

May 1, 2026

1. In the figure to the right, does curve A lead or lag curve B?

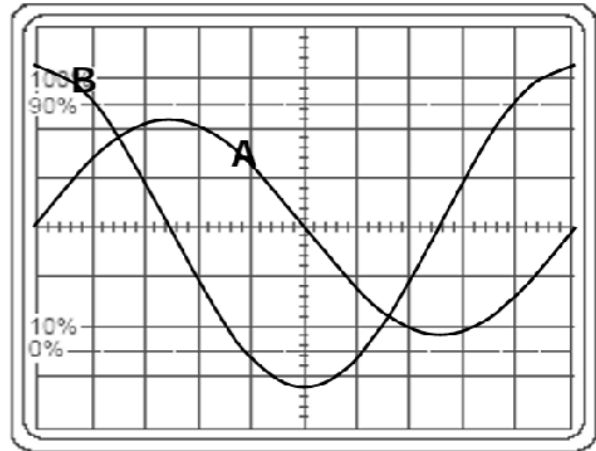
Curve B leads curve A since B reaches its peak before curve A does.

2. The oscilloscope vertical scale is set to 5 V/div and the horizontal scale is set to 5 ms/div.

- a) What is the time difference in the curves?

Curve A reaches its peak approximately 2.5 divisions after curve B does, so:

$$\Delta t \approx (2.5 \text{ div})(5 \text{ ms/div}) = \mathbf{12.5 \text{ ms.}}$$



- b) What is the phase difference in the curves?

Assuming both curves have the same frequency, curve B leads A by approximately:

$$\Delta\phi = \mathbf{90^\circ = \pi/2 \text{ radians.}}$$

This is because curve B reaches its peak approximately a quarter of a period before curve A does. More precisely, if $t = 0$ was fixed on the left side of the screen, curve B roughly models $B_0 \cos(\omega t)$ whereas A models $A_0 \sin(\omega t) = A_0 \cos(\omega t - \pi/2)$.

3. A resistor, inductor, and capacitor are connected in series and the potential difference across each are measured as $V_R = 35 \text{ V}$, $V_L = 45 \text{ V}$, and $V_C = 55 \text{ V}$ respectively. What is the input voltage V ?

The impedance of the circuit is given by:

$$Z = \sqrt{R^2 + (X_L - X_C)^2}.$$

By “Ohm’s law for impedance”, the potential difference across the LRC components is given by:

$$\begin{aligned} \sum V &= IZ = I\sqrt{R^2 + (X_L - X_C)^2} \\ &= \sqrt{I^2 R^2 + I^2 (X_L - X_C)^2} \\ &= \sqrt{(IR)^2 + (IX_L - IX_C)^2} \\ &= \sqrt{V_R^2 + (V_L - V_C)^2}. \end{aligned}$$

By Kirchhoff’s voltage law, the input voltage V must be equal to the sum of the potential differences across the components, so:

$$V = \sqrt{V_R^2 + (V_L - V_C)^2} = \sqrt{(35 \text{ V})^2 + (45 \text{ V} - 55 \text{ V})^2} = \mathbf{36.4 \text{ V.}}$$